Healthcare Analytics Dashboard

Project Overview

This project demonstrates an end-to-end Healthcare Analytics Dashboard built using Microsoft Power BI. It transforms synthetic healthcare data into interactive insights for hospital executives, clinicians, pharmacy managers, and finance teams.

Business Objective

Develop an interactive dashboard that supports healthcare decision-making by analyzing patient demographics, disease burden, medication utilization, clinical observations, and financial performance.

Dataset

Source: Synthea Synthetic Healthcare Dataset.
Tables Used: Patients, Encounters, Medications, Conditions, Observations, and a custom Calendar table.

Data Model

A relational data model connects patients, encounters, medications, conditions, and observations to support interactive filtering and cross-table analysis.

Dashboard Pages

1. Executive Summary
Hospital KPIs, county analysis, and trends.

2. Pharmacy Analytics
Medication utilization, Top-N analysis, encounter class, and prescription insights.

3. Clinical & Population Analytics
Disease burden, demographics, observations, and clinical cost analysis.

4. Financial Analytics
Insurance coverage, patient payments, financial burden, and expenditure analysis.

Key DAX Concepts

CALCULATE(), FILTER(), ALL(), VALUES(), SUMX(), AVERAGEX(), RANKX(), TOPN(), SELECTEDVALUE(), DIVIDE(), and VAR.

Business Insights

• Identify high-cost medications and diseases.
• Compare insurance coverage with patient out-of-pocket costs.
• Monitor disease burden across demographics.
• Support strategic healthcare decision-making.

Tools & Technologies

Microsoft Power BI, PostgreSQL, DAX, Power Query, GitHub, and Synthea.

Future Enhancements

Integration with real Electronic Health Records, predictive analytics, real-time refresh, and row-level security.